

# Detailed Field-Geometric Spectral Theory

Full Derivation of Particle Spectra, Mass Hierarchies  
and Gauge Groups from the  $T^4/\mathbb{Z}_3$  Geometry

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**Correction notes**

**Corrections from the calculation review (1 Oct. 2026).** The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Higgs route to  $\xi$  (note):  $1/(16\pi^3) = 2.016 \cdot 10^{-3}$  is 15 times  $4/30000$ ; the iteration  $x \mapsto x(1 - 100x)$  has only the fixed point 0; the "second independent route" is not established. With  $\lambda_h = m_h^2/(2v^2)$  a consistency check to about 2.3 % remains (Doc. 354, A190, Doc. 385). Cross-reference table: " $\xi$  from Higgs geometry (second route), Doc. A130"  $\rightarrow$  "Higgs consistency check of  $\xi$  ( $\approx 2.3\%$ ), Doc. 354, A190" (A130 does not contain the Higgs route, Doc. 133 contains no iteration).
- Exponent ladder: "arithmetic, step  $-1/3$ "  $\rightarrow$  geometric,  $p_g = (3/2)(2/3)^{g-1}$ .
- $m_\tau = 1783.4$  MeV (+0.37 %) instead of 1783.5 MeV (+0.38 %).
- $m_e$ : 0.511 MeV,  $< 0.1\%$   $\rightarrow$  0.50499 MeV,  $-1.18\%$  (three places).
- Neutrino note:  $\Delta m_{\text{atm}}^2 +13.8\%$   $\rightarrow$   $+16.0\%$  (with  $K_{\text{frak}} +12.8\%$  instead of  $+12.9\%$ ).
- $W^\pm, Z^0$ : 80.38/91.19 GeV are inputs (measured  $g, \theta_W$ ), not a [K] prediction.

**List of Symbols**

| Symbol      | Meaning                              | Value / Unit                          |
|-------------|--------------------------------------|---------------------------------------|
| $\xi$       | FFGFT geometry parameter             | $4/30000 = 1.3\bar{3} \times 10^{-4}$ |
| $v$         | Higgs vacuum expectation value       | 246.0 GeV                             |
| $m_e$       | electron mass (neutrino sector ref.) | 0.511 MeV                             |
| $r_i$       | topological winding ratio            | $n_\phi^2/n_\theta$                   |
| $p_i$       | exponent in mass formula             | dimensionless                         |
| $\eta_{D4}$ | $D_4$ projection coefficient         | 81/640                                |
| $F_{D4}$    | $D_4$ volume element                 | 27/1 600 000                          |
| [K]         | derivable from $\xi$                 | status marker                         |
| [B]         | algebraically proved                 | status marker                         |
| [S]         | sketched / plausible                 | status marker                         |

**Summary**

The FFGFT derives all particle masses of the leptonic sector from the single dimensionless geometry parameter  $\xi = 4/30000$  and the topology of the  $T^4/\mathbb{Z}_3$  orbifold. This document follows the derivation explicitly in every step: from the sub-Planck lattice, through the  $D_4$  projection and topological winding numbers, to the measurable masses. The mathematical Hilbert-space foundation is provided in Doc. 322.

**Status markers**

All statements carry FFGFT status markers:

**[K]** derivable from  $\xi$    **[B]** algebraically proved   **[S]** sketched/plausible   **[SETZUNG]** axiomatic setting

## .1 Starting Point: the Sub-Planck Geometry

### Why a sub-Planck level?

The Standard Model describes particle masses through Yukawa couplings – free parameters measured experimentally but not explained within the framework itself. FFGFT goes one level deeper: masses do not originate from free couplings but from the geometry of space-time on scales below the Planck length  $\ell_P$ .

**[SETZUNG]** At the sub-Planck scale  $t_0$  space-time has a discrete packing structure. The most efficient regular lattice in four dimensions is the  $D_4$  lattice – the 4D analogue of the densest sphere packing in 3D.

### The $D_4$ lattice and its symmetry

Properties of the  $D_4$  lattice:

- **4 orthogonal dimensions** with identical scaling.
- **Automorphism group**  $\text{Aut}(D_4) \cong W(F_4)$  with  $|W(F_4)| = 1152$  elements.
- **24 nearest neighbours** per lattice point.
- The torus  $T^4/\mathbb{Z}_3$  is the natural compactification, with the  $\mathbb{Z}_3$  identification reflecting the three quark colour charges.

**[K]** The fundamental FFGFT relation  $\tilde{T} \cdot m = 1$  follows from the duality between torus radius  $R \propto \tilde{T}$  and mass  $m \propto 1/R$ : on a compact torus the smallest mode spacing is inverse to the radius.

## .2 The Geometry Parameter $\xi$ and its Origin

### Definition and numerical value

**[K]**

$$\xi = \frac{4}{30000} = \frac{1}{7500} = 1.3\bar{3} \times 10^{-4}. \quad (1)$$

**Numerator 4** equals the number of orthogonal dimensions of the  $D_4$  lattice – topologically fixed, not freely chosen.

**Denominator 30000** equals the phase volume of the fractal  $T^4$  compactification: product of the  $\mathbb{Z}_3$  orbifold order (3) and the number of fundamental Fourier modes on the torus ( $10^4$ , the fourth power of the first non-trivial mode).

### Connection to Higgs geometry

**[K]** An independent derivation of  $\xi$  follows from the Higgs mass (Doc. A130):  $\xi = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$ . With the  $K_{\text{frac}}$ -interface factor and iteration the fixed point  $\xi = 4/30000$  is obtained (Doc. 133). The Higgs path and the  $D_4$ -lattice path are two independent routes to the same  $\xi$ . (Note of 1 Oct. 2026: with  $\lambda_h = m_h/v$  (an insertion found only in the German version) the formula gives  $1/(16\pi^3) = 2.016 \cdot 10^{-3}$ , 15 times  $4/30000$ . The iteration  $\xi_{n+1} = \xi_n(1 - 100\xi_n)$  runs from  $2.016 \cdot 10^{-3}$  via  $1.61 \cdot 10^{-3}$ ,  $1.35 \cdot 10^{-3}$ , ... towards 0 (its only fixed point), not to  $4/30000$ ; the reading  $\xi = \xi_0(1 - 100\xi)$  also gives  $1.68 \cdot 10^{-3}$ . The Higgs path therefore does not lead to  $4/30000$  in this form. The error lies in inserting  $\lambda_h = m_h/v$  with the Standard Model self-coupling  $\lambda_h = m_h^2/(2v^2) = 0.129$

one has  $\xi_{EFT} = m_h^2/(64\pi^3 v^2) = 1.30 \cdot 10^{-4}$ , about 2.3 % below  $4/30000$  (Doc. 354, A190, Doc. 385; R59). The Higgs geometry thus remains a consistency check of  $\xi$  to within a few per cent, not an exact derivation. Doc. A130 does not contain the Higgs route, Doc. 133 contains no such iteration.)

### The $D_4$ projection coefficient $\eta_{D4}$

**[B]** The projection coefficient:

$$\eta_{D4} = \frac{81}{640} = 0.1265625, \quad (2)$$

and the  $D_4$  volume element:

$$F_{D4} = \xi \cdot \eta_{D4} = \frac{27}{1\,600\,000} = 1.6875 \times 10^{-5}. \quad (3)$$

Numerical check:  $4 \cdot 81 = 324$ ,  $30000 \cdot 640 = 19\,200\,000$ ,  $324/19\,200\,000 = 27/1\,600\,000$ . ✓

## .3 Charged Leptons: Derivation from Winding Numbers

### Why winding numbers?

Field modes on a compact torus  $T^2$  are standing waves. The periodicity condition  $\psi(x+2\pi R) = \psi(x)$  permits only integer winding numbers  $n \in \mathbb{Z}$ . On the  $T^2$  torus two independent circles carry winding numbers  $n_\theta$  and  $n_\phi$ .

**[K]** The three generations of charged leptons correspond to the three topologically distinguished  $(n_\theta, n_\phi)$  states forming the simplest rational ratios  $r_i = n_\phi^2/n_\theta$ :

**Table 1:** FFGFT quantum numbers of the charged leptons

| Particle     | $n_\theta$ | $n_\phi$ | $r_i = n_\phi^2/n_\theta$ | $p_i$ |
|--------------|------------|----------|---------------------------|-------|
| Electron $e$ | 3          | 2        | 4/3                       | 3/2   |
| Muon $\mu$   | 5          | 4        | 16/5                      | 1     |
| Tau $\tau$   | 9          | 5        | 25/9                      | 2/3   |

### The exponent ladder

**[K]** The exponents  $p_i$  form a geometric sequence with ratio  $2/3$  per generation,  $p_g = \frac{3}{2} \left(\frac{2}{3}\right)^{g-1}$ :

$$p_e = \frac{3}{2}, \quad p_\mu = 1, \quad p_\tau = \frac{2}{3}. \quad (4)$$

(Corrected on 1 Oct. 2026: previously “an arithmetic ladder with step  $\Delta p = -1/3$  per generation – a direct imprint of the  $\mathbb{Z}_3$  orbifold order” – the steps  $3/2 \rightarrow 1 \rightarrow 2/3$  are  $-1/2$  and  $-1/3$ ; the constant quantity is the ratio  $2/3$ .) The general mass formula:

$$m_i = r_i \cdot \xi^{p_i} \cdot v, \quad v = 246.0 \text{ GeV}. \quad (5)$$

## Step-by-step mass calculation

Pre-computed  $\xi$  powers:

$$\xi^{3/2} = 1.5396 \times 10^{-6}, \quad \xi^1 = 1.3\bar{3} \times 10^{-4}, \quad \xi^{2/3} = 2.6099 \times 10^{-3}. \quad (6)$$

[K]

$$m_e = \frac{4}{3} \cdot 1.5396 \times 10^{-6} \cdot 246.0 \times 10^3 \text{ MeV} = \mathbf{0.50499 \text{ MeV}} \quad (\text{PDG: } 0.511 \text{ MeV}, -1.18 \%), \quad (7)$$

$$m_\mu = \frac{16}{5} \cdot 1.3\bar{3} \times 10^{-4} \cdot 246.0 \times 10^3 \text{ MeV} = \mathbf{104.96 \text{ MeV}} \quad (\text{PDG: } 105.66 \text{ MeV}), \quad (8)$$

$$m_\tau = \frac{25}{9} \cdot 2.6099 \times 10^{-3} \cdot 246.0 \times 10^3 \text{ MeV} = \mathbf{1783.4 \text{ MeV}} \quad (\text{PDG: } 1776.86 \text{ MeV}). \quad (9)$$

## Parameter-free mass ratios

[K] The VEV  $v$  cancels in ratios:

$$m_\mu/m_e = \frac{12}{5} \cdot \xi^{-1/2} = 207.85 \quad (\text{PDG: } 206.77, +0.52 \%), \quad (10)$$

$$m_\tau/m_\mu = \frac{125}{144} \cdot \xi^{-1/3} = 16.99 \quad (\text{PDG: } 16.82, +1.04 \%), \quad (11)$$

$$m_\tau/m_e = \frac{25}{12} \cdot \xi^{-5/6} = 3532 \quad (\text{PDG: } 3477, +1.57 \%). \quad (12)$$

**Table 2:** Lepton masses and ratios: FFGFT vs. PDG

| Quantity       | Formula                     | Theory      | PDG         | Dev.    | Status |
|----------------|-----------------------------|-------------|-------------|---------|--------|
| $m_e$          | $\frac{4}{3}\xi^{3/2}v$     | 0.50499 MeV | 0.511 MeV   | -1.18 % | [K]    |
| $m_\mu$        | $\frac{16}{5}\xi^1v$        | 104.96 MeV  | 105.66 MeV  | -0.66 % | [K]    |
| $m_\tau$       | $\frac{25}{9}\xi^{2/3}v$    | 1783.4 MeV  | 1776.86 MeV | +0.37 % | [K]    |
| $m_\mu/m_e$    | $\frac{12}{5}\xi^{-1/2}$    | 207.85      | 206.77      | +0.52 % | [K]    |
| $m_\tau/m_\mu$ | $\frac{125}{144}\xi^{-1/3}$ | 16.99       | 16.82       | +1.04 % | [K]    |

## .4 Neutrino Sector: Fixed-Point Localisation on the Orbifold

**Note (31 August 2026):** This section derives neutrino masses from orbifold fixed points ( $F_2, F_5, F_7$ , exponents  $\{9/4, 2, 9/5\}$ ) and achieves  $\Delta m_{\text{atm}}^2$  with a deviation of +16.0 % (with  $K_{\text{frak}} +12.8\%$ ; bridge R87) (*Corrected on 1 Oct. 2026: previously +13.8 % – within this document*  $2845.4/2453 - 1 = +16.0\%$ , with  $K_{\text{frak}} 2767.8/2453 - 1 = +12.8\%$ ; +13.8 % would only result against  $2.50 \times 10^{-3} \text{ eV}^2$ ). Doc. 339/340 (August 2026) derives both mass differences from the  $GF(27)^*$  Frobenius orbit structure and closes R87 at 1.0 % deviation [K]. The present fixed-point derivation is retained as a historical development step; the current derivation is Doc. 340.

**Addendum (3 September 2026, Doc. 343 Theorem D''')**: The quadratic character mod 13 (self-inverseness of the Frobenius orbit under  $k \mapsto k^{-1}$ ) determines Majorana/Dirac for the Galois masses in Doc. 340:  $\nu_1$  (orbit  $O_1$ ) and  $\nu_2$  (orbit  $O_3$ ) are Majorana,  $\nu_3$  (orbit  $O_4$ ) is a **Dirac neutrino [B]**. The fixed-point assignment  $F_2, F_5, F_7$  in this section provides no Majorana/Dirac character; this follows only from the Galois algebra of Doc. 340/343.

## Structural difference from charged leptons

Charged leptons arise as winding modes on the  $T^2$  torus and couple to the Higgs VEV  $v$ . Neutrinos are fundamentally different: they are localised at the *geometric fixed points* of the  $T^4/\mathbb{Z}_3$  orbifold – points  $x_0$  satisfying  $\tau(x_0) = x_0$ . This topological distinction produces a double  $\xi$ -suppression compared to the charged leptons. For the formal Hilbert-space treatment of the fixed-point boundary conditions see Doc. 322, Eq. (15).

## The neutrino mass formula

[K]

$$m_{\nu_i} = m_e \cdot \xi^{p_i}, \quad m_e = 0.511 \text{ MeV}. \quad (13)$$

**Table 3:** Fixed-point localisation of neutrino flavours (Doc. 322)

| Fixed pt. | Position                       | Neutrino | Exponent $p_i$ | Mass      |
|-----------|--------------------------------|----------|----------------|-----------|
| $F_2$     | $\frac{2\pi R}{3}(1, 1, 1, 0)$ | $\nu_1$  | $9/4$          | 0.976 meV |
| $F_5$     | $\frac{4\pi R}{3}(1, 1, 1, 0)$ | $\nu_2$  | 2              | 9.084 meV |
| $F_7$     | $\frac{4\pi R}{3}(1, 1, 1, 1)$ | $\nu_3$  | $9/5$          | 54.11 meV |

## Step-by-step neutrino mass calculation

[K] Pre-computed  $\xi$  powers and masses:

$$\xi^{9/4} = 1.9103 \times 10^{-9}, \quad m_{\nu_1} = 511\,000 \text{ eV} \cdot 1.9103 \times 10^{-9} = \mathbf{0.976 \text{ meV}}, \quad (14)$$

$$\xi^2 = 1.7 \times 10^{-8}, \quad m_{\nu_2} = 511\,000 \text{ eV} \cdot 1.7 \times 10^{-8} = \mathbf{9.084 \text{ meV}}, \quad (15)$$

$$\xi^{9/5} = 1.059 \times 10^{-7}, \quad m_{\nu_3} = 511\,000 \text{ eV} \cdot 1.059 \times 10^{-7} = \mathbf{54.11 \text{ meV}}. \quad (16)$$

## Mass-squared differences and oscillation dynamics

### Solar mass-squared difference $\Delta m_{21}^2$

$$\Delta m_{21}^2 = (9.084 \text{ meV})^2 - (0.976 \text{ meV})^2 = 82.52 - 0.95 = 81.57 \text{ meV}^2 = \mathbf{8.16 \times 10^{-5} \text{ eV}^2}. \quad (17)$$

PDG 2023:  $7.53 \times 10^{-5} \text{ eV}^2$ . Deviation: +8.3 %.

### Atmospheric mass-squared difference $\Delta m_{32}^2$

$$\Delta m_{32}^2 = (54.11 \text{ meV})^2 - (9.084 \text{ meV})^2 = 2927.9 - 82.5 = 2845.4 \text{ meV}^2 = \mathbf{2.845 \times 10^{-3} \text{ eV}^2}. \quad (18)$$

PDG (NuFIT 5.3):  $2.453 \times 10^{-3} \text{ eV}^2$ . Deviation: +16.0 %.

## Open point: atmospheric mass-squared difference

$\Delta m_{21}^2$  agrees to +8.3 %, within the typical FFGFT range.  $\Delta m_{32}^2$  deviates by +16.0 % – about twice as much and with the same sign:  $m_{\nu_3}$  is too large. Equal sign and comparable magnitude point to a common cause rather than two independent problems.

The fixed-point exponents  $p_i$  are secured [K]; the source of the discrepancy is an open point. Three candidates were examined:

- **$K_{\text{frak}}$  correction at fixed point  $F_7$**  (Doc. 315):  $F_7$  is the only fixed point in the full (1, 1, 1, 1) direction and thus carries a mass-circle component. With  $K_{\text{frak}} = 1 - 100\xi = 74/75$  one obtains  $m_3 = 53.39$  meV and  $\Delta m_{32}^2 = 2.768 \times 10^{-3} \text{ eV}^2$ , i.e. +12.8 % – an improvement, but not sufficient.
- **Mixing term  $F_5$ – $F_7$** : the fixed points are not fully orthogonal. An off-diagonal term in the mass matrix on the orbifold would shift both eigenvalues and would carry the right sign. This requires the complete mass matrix, not worked out here – the most promising candidate.
- **A different exponent** – unlikely: the exponent required for the PDG value is  $p_3 = 1.8081$ , only 0.45 % above  $9/5 = 1.8000$ . No rational fraction with small numerators lies closer ( $16/9 \rightarrow +74\%$ ,  $11/6 \rightarrow -37.5\%$ ,  $20/11 \rightarrow -17.1\%$ ). The exponent  $9/5$  is the best rational candidate and is grounded in the  $D_4$  coordination.

The deviation therefore most likely does not stem from a wrong exponent but from an effect correcting  $m_{\nu_i} = m_e \xi^{p_i}$  multiplicatively. Status: [S].

## Cosmological mass sum

$$\sum_i m_{\nu_i} = 0.976 + 9.084 + 54.11 \text{ meV} = 64.17 \text{ meV} = 0.0642 \text{ eV}. \quad (19)$$

Planck 2018 limit:  $\sum m_\nu < 0.12 \text{ eV}$ . ✓

**Table 4:** Neutrino spectrum: FFGFT prediction vs. experiment

| Quantity          | Formula                     | Prediction                          | Experiment                          | Status      |
|-------------------|-----------------------------|-------------------------------------|-------------------------------------|-------------|
| $m_{\nu_1}$       | $m_e \xi^{9/4}$             | 0.976 meV                           | $< 0.8 \text{ eV}$                  | [K]         |
| $m_{\nu_2}$       | $m_e \xi^2$                 | 9.084 meV                           | $\sim 8.7 \text{ meV}$              | [K]         |
| $m_{\nu_3}$       | $m_e \xi^{9/5}$             | 54.11 meV                           | $\sim 50 \text{ meV}$               | [K]         |
| $\Delta m_{21}^2$ | $m_{\nu_2}^2 - m_{\nu_1}^2$ | $8.16 \times 10^{-5} \text{ eV}^2$  | $7.53 \times 10^{-5} \text{ eV}^2$  | [K] +8.3 %  |
| $\Delta m_{32}^2$ | $m_{\nu_3}^2 - m_{\nu_2}^2$ | $2.845 \times 10^{-3} \text{ eV}^2$ | $2.453 \times 10^{-3} \text{ eV}^2$ | [S] +16.0 % |
| $\sum m_\nu$      | $\sum m_{\nu_i}$            | 0.0642 eV                           | $< 0.12 \text{ eV}$                 | [K]         |

## .5 Symmetry Breaking and Gauge Structure

### $SU(3)_c$ : colour charges from $\mathbb{Z}_3$ fibration

**[K]** The  $\mathbb{Z}_3$  symmetry generates three discrete complex-conjugate phase levels  $\omega = e^{2\pi i/3}$ , corresponding to the three colour charges Red, Green, Blue. The eight generators of  $\mathfrak{su}(3)$  follow from the eight real degrees of freedom of the  $\mathbb{Z}_3$  projection. **[B]** Full algebraic derivation of the Gell-Mann matrices from orbifold modes is in Doc. 321.

## Electroweak bosons ( $W^\pm, Z^0$ )

[B]

$$m_W = \frac{1}{2} g v \approx 80.38 \text{ GeV}, \quad m_Z = m_W / \cos \theta_W \approx 91.19 \text{ GeV}. \quad (20)$$

(Note of 1 Oct. 2026: the numerical values insert the measured  $g$  and  $\theta_W$  and are inputs, not a prediction: with  $g = 0.6517$ ,  $v = 246 \text{ GeV}$  one gets  $m_W = 80.16 \text{ GeV}$ ,  $80.38 \text{ GeV}$  requires  $g = 0.6535$ ;  $m_Z = m_W / \cos \theta_W$  with  $\cos \theta_W = m_W / m_Z$  is circular.)

## .6 Complete Overview of All Spectra

**Table 5:** Systematic overview of all FFGFT spectra, Doc. 320

| Group        | Particle          | Prediction                          | PDG                                 | Status      |
|--------------|-------------------|-------------------------------------|-------------------------------------|-------------|
| Leptons      | Electron $e$      | 0.50499 MeV                         | 0.511 MeV                           | [K]         |
|              | Muon $\mu$        | 104.96 MeV                          | 105.66 MeV                          | [K]         |
|              | Tau $\tau$        | 1783.4 MeV                          | 1776.86 MeV                         | [K]         |
| Neutrinos    | $\nu_1$ ( $F_2$ ) | 0.976 meV                           | $< 0.8 \text{ eV}$                  | [K]         |
|              | $\nu_2$ ( $F_5$ ) | 9.084 meV                           | $\sim 8.7 \text{ meV}$              | [K]         |
|              | $\nu_3$ ( $F_7$ ) | 54.11 meV                           | $\sim 50 \text{ meV}$               | [K]         |
| Oscillations | $\Delta m_{21}^2$ | $8.16 \times 10^{-5} \text{ eV}^2$  | $7.53 \times 10^{-5} \text{ eV}^2$  | [K] +8.3 %  |
|              | $\Delta m_{32}^2$ | $2.845 \times 10^{-3} \text{ eV}^2$ | $2.453 \times 10^{-3} \text{ eV}^2$ | [S] +16.0 % |
|              | $\sum m_\nu$      | 0.0642 eV                           | $< 0.12 \text{ eV}$                 | [K]         |
| Bosons       | $\gamma, g$       | 0                                   | 0                                   | [K]         |
|              | $W^\pm$           | 80.38 GeV (input)                   | 80.38 GeV                           | input       |
|              | $Z^0$             | 91.19 GeV (input)                   | 91.19 GeV                           | input       |

(Corrected on 1 Oct. 2026:  $W^\pm$  and  $Z^0$  rows previously listed as a prediction with [K] – the values insert the measured  $g$  and  $\theta_W$  ( $\frac{1}{2} \cdot 0.6517 \cdot 246 \text{ GeV} = 80.16 \text{ GeV}$ ),  $\theta_W$  is not derived from  $\xi$  here (cf. Doc. 323); they are inputs.)

## .7 Cross-references and Open Bridges

**Table 6:** Derivation chain Doc. 320 and cross-references

| Step                                                 | Doc.           | Status      |
|------------------------------------------------------|----------------|-------------|
| $\xi = 4/30000$ from $D_4$ lattice                   | Docs. 001, 205 | [K]         |
| Higgs consistency check of $\xi$ ( $\approx 2.3\%$ ) | Doc. 354, A190 | [K]         |
| $K_{\text{frac}}$ interface factor                   | Docs. 012, 133 | [K]         |
| $\tilde{T} \cdot m = 1$                              | Doc. 003       | [SETZUNG]   |
| Winding numbers $(n_\theta, n_\phi)$                 | Doc. 317       | [K]         |
| Lepton masses $m_e, m_\mu, m_\tau$                   | Docs. 006, 317 | [K]         |
| Hilbert-space embedding, operator $\hat{F}$          | Doc. 322       | [K/S]       |
| Fixed-point positions $F_2, F_5, F_7$                | Doc. 322       | [K]         |
| $\Delta m_{21}^2$ (+8 %)                             | this Doc.      | [K]         |
| $\Delta m_{32}^2$ (+16.0 %)                          | this Doc.      | [S] open    |
| $SU(3)_c$ from $\mathbb{Z}_3$                        | Doc. 321       | [B]         |
| Weinberg angle $\theta_W$ from $\xi$                 | Doc. 323       | [K]         |
| Quark sector (hadrons)                               | Docs. 318, 319 | open bridge |

(Corrected on 1 Oct. 2026: row previously " $\xi$  from Higgs geometry (second route) & Doc. A130 & [K]" – as a derivation the Higgs path does not lead to  $4/30000$ ; as a consistency check it matches  $\xi$  to about 2.3 % (Doc. 354, A190, Doc. 385), see the note in the section "Connection to Higgs geometry". Doc. A130 does not contain the Higgs route; Doc. 133 contains no iteration.)